YIYOU SUN

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Research Overview

My research develops principled algorithms and evaluations for models to **be aware of**, **learn from**, and **generalize to** Out-of-distribution (OOD) inputs across task settings ranging from image classification to prompt-based reasoning in modern LLMs. My current research particularly focuses on the following areas:

OOD Learnability: How can we enable LLMs to solve previously unsolvable tasks using RLVR? [4]

OOD Generalization: In which aspects can inference-time reasoning be generalized in mathematical problems, programming code, and agentic tasks? [6]

Academic Experience

University of California, Berkely Aug 2024 - Aug 2026
Postdoctoral Researcher *Berkeley, CA*

- Advised by Prof. Dawn Song.
- Affiliated with Simons Institute for the Theory of Computing.

University of Wisconsin-Madison Sept 2020 - July 2023
Research Assisstant *Madison, WI*

- Advised by Prof. Sharon (Yixuan) Li. (2020 - 2023)
- Advised by Prof. Vikas Singh. (2018 - 2020)

MIT: Computer Science and Artificial Intelligence Laboratory Dec 2017 - May 2018
Undergraduate Research Assisstant *Cambridge, MA*

- Supervised by Prof. Bolei Zhou (now at UCLA) and Prof. Antonio Torralba.

Harvard University: DASlab July 2017 - May 2018
Undergraduate Research Assisstant *Cambridge, MA*

- Supervised by Prof. Stratos Idreos.

Education

University of Wisconsin-Madison Sept. 2018 - July. 2023
PhD. in Computer Science, Minor in Math *Madison, US*

- Thesis "Detecting and Learning Out-of-distribution Data in the Open-world: Algorithm and Theory"

Harbin Institute of Technology Sept. 2014 - Jun. 2018
B.S. in Computer Science *Harbin, China*

- GPA 93.67/100. Rank **1st** among 97 students.

Industrial Experience

NEC Laboratories America Inc. Aug. 2023 - Aug. 2024
Researcher *Plainsboro, NJ*

Google Research May 2022 - Sept 2022
Research Intern *Mountain View, CA*

Meta AI June 2020 - Sept 2020 & June 2021 - Sept 2021
Research Intern *Remote*

Press

Data Science Collective (~1M followers) 2025
The AI Reasoning Ladder: What Machines Can Do — And Where They Still Fail

AI Era (新智元, Top 3 Chinese AI Media) 2025
RL breaks through the "cognitive limit" of the model, and true learning happens!

Medium (~1K likes) 2025
OMEGA Math Challenge, Testing AI's Reasoning Edge

Research Blogs

BerkeleyRDI	RL Grokking Recipe – How Can We Enable LLMs to Solve Previously Unsolvable Tasks?	2025
AI2	Can LLMs Reason Outside the Box in Math?	2025
LESSWRONG	Can Aha Moments be Fake? Identifying True and Decorative Thinking Steps in CoT	2025

Students Mentoring

Longfei Ma, PhD student at the Zhejiang University 2023 - 2024
1 paper at NeurIPS 2024

Xue Li, PhD candidate at the University of Wisconsin, Madison 2024
1 paper at ICLR 2025

Haoyu Bai, PhD candidate at the University of Wisconsin, Madison 2024 - 2025
1 paper at CVPR 2025, 1 paper in submission

Georgia Zhou, Undergraduate student at the University of California, Berkeley 2025
1 paper at NeurIPS 2025, 1 paper at NeurIPS Workshop (MATH-AI) 2025

Ken Zheng, Undergraduate student at the University of California, Berkeley 2025
1 paper at NeurIPS 2025

Jiachen Zhao, PhD student at the Northeastern University 2025
MATS scholar fellowship, 1 paper in submission

Weichen Zhang, Undergraduate student at the Tongji University 2025
1 paper in submission

Selected Talks

RL Grokking Recipe: How Does RL Unlock and Transfer New Algorithms in LLMs? Nov 2025
Invited Talk @ UW-Madison Machine Learning Research Group *Madison, WI*

The Theoretical Foundation for the Open-world Representation Learning: A Spectral View Nov 2023
Invited Talk @ UIUC ML Seminar *Remote*

Out-of-distribution (OOD) Generalization and Detection in AI Risk Management Oct 2023
Invited Talk @ Beijing Academy of Artificial Intelligence *Remote*

Detecting and Learning Out-of-Distribution Data in the Open world July 2023
Ph.D. Oral Defense *Madison, WI*

Rethinking Domain Generalization for Face Anti-spoofing: Separability and Alignment Sept 2022
Presentation @ Google Research *Mountain View, CA*

Teaching

LLM Agent is Hallucinating and Where to Find Them? Feb 2025
Research Track Lead @ UC Berkeley CS194/294-280 Advanced Large Language Model Agents *Berkeley, CA*

Introduction on Trustworthy Machine Learning Nov 2023
Guest lecture @ UC Riverside CS/ECE 228 – Introduction to Deep Learning *Riverside, CA*

Introduction on Interpretable Deep Learning Oct 2020
Guest lecture @ UW Madison CS 839 – Special Topics in AI: Deep Learning *Madison, WI*

Community Services

Chair

- Workshop Session in Agentic AI Summit - Berkeley RDI in 2025

Co-organizer

- Socially Responsible and Trustworthy Foundation Models (ResponsibleFM) in NeurIPS 2025
- ML Alignment & Theory Scholars (MATS) Mentorship Program in Summer 2025

Journal Reviewer

- Review for Nature Machine Intelligence in 2025.
- Review for IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) in 2024.
- Review for IEEE Transactions on Multimedia in 2023.
- Review for IEEE Transactions on Image Processing (TIP) in 2023.
- Review for International Journal of Computer Vision (IJCV) in 2022, 2023.
- Review for IEEE Transactions on Cybernetics in 2022.
- Review for Pattern Recognition in 2021, 2022.

Conference Reviewer

- Review for the International Conference on Artificial Intelligence and Statistics (AISTATS) in 2025.
- Review for the International Conference on Computer Vision and Pattern Recognition (CVPR) in 2022, 2023, 2024.
- Review for International Conference on Machine Learning (ICML) in 2022 - 2026.
- Review for Conference on Neural Information Processing Systems in 2022 - 2026.
- Review for European Conference on Computer Vision (ECCV) in 2022.
- Review for International Conference on Learning Representations (ICLR) in 2022 - 2026.
- Review for Association for the Advancement of Artificial Intelligence (AAAI) in 2022, 2023.

Awards and Honors

Most Influential ICML Papers (ICML 2022, #12 cited) [link] 2025
Best Reviewer Award (4%) at AISTATS 2025
Top Reviewer Award (6%) at NeurIPS 2022
National Scholarship (0.2%) from Ministry of Education of the People's Republic of China 2018
Outstanding Award (Rank 1st) at the 9th Intel Cup National Collegiate Software Innovation Competition 2017
National Scholarship (0.2%) from Ministry of Education of the People's Republic of China 2017
National Scholarship (0.2%) from Ministry of Education of the People's Republic of China 2016

Publications / Preprints / Patents (Sorted by Years)

(* indicates equal contribution, **grey** color indicates the core mentorship role)

[1] Haoyue Bai, **Yiyou Sun**, Wei Cheng, and Haifeng Chen. "Where's the Liability in the Generative Era? Recovery-based Black-Box Detection of AI-Generated Content". in: *Computer Vision and Pattern Recognition*, 2025.

[2] Xue Li*, **Yiyou Sun***, Wei Cheng, Yinglun Zhu, and Haifeng Chen. "Chain-of-region: Visual language models need details for diagram analysis". in: *International Conference on Learning Representations*, 2025.

[3] Jacob Light, Yu Wu, **Yiyou Sun**, Wen Yu, Xiang Zhao, Zhangyang Hu, Haifeng Chen, and Wei Cheng. "Scattered Forest Search: Smarter Code Space Exploration with LLMs". in: *International Conference on Learning Representations*, 2025.

[4] **Yiyou Sun**, Yuhan Cao, Pohao Huang, Haoyue Bai, Hannaneh Hajishirzi, Nouha Dziri, and Dawn Song. *RL Grokking Recipe: How Does RL Unlock and Transfer New Algorithms in LLMs?* arXiv:2509.21016. 2025.

[5] **Yiyou Sun**, Yu Gai, Lijie Chen, Abhilasha Ravichander, Yejin Choi, and Dawn Song. "Why and How LLMs Hallucinate: Connecting the Dots with Subsequence Associations". in: *Advances in Neural Information Processing Systems*, 2025, (**Cited by Kalai et al., OpenAI**).

[6] **Yiyou Sun**, Shawn Hu, Georgia Zhou, Ken Zheng, Hannaneh Hajishirzi, Nouha Dziri, and Dawn Song. "OMEGA: Can LLMs Reason Outside the Box in Math? Evaluating Exploratory, Compositional, and Transformative Generalization". in: *Advances in Neural Information Processing Systems*, 2025.

[7] **Yiyou Sun**, Georgia Zhou, Hao Wang, Dacheng Li, Nouha Dziri, and Dawn Song. "Climbing the Ladder of Reasoning: What LLMs Can-and Still Can't-Solve after SFT?" In: *Advances in Neural Information Processing Systems Workshop (MATH-AI)*, 2025.

[8] Yifan Zhang, Xiang Liu, **Yiyou Sun**, Abdullah Alharbi, Hana Alzahrani, Badr Alomair, and Dawn Song. *Can LLMs Design Good Questions Based on Context?* arXiv:2501.03491. 2025.

[9] Weichen Zhang*, **Yiyou Sun***, Pohao Huang, Jiayue Pu, Heyue Lin, and Dawn Song. *MIRAGE-Bench: LLM Agent is Hallucinating and Where to Find Them.* arXiv:2507.21017. 2025.

[10] Qiwei Zhao, Xujiang Zhao, Yanchi Liu, Wei Cheng, **Yiyou Sun**, Mika Oishi, Takao Osaki, Katsushi Matsuda, Huaxiu Yao, and Haifeng Chen. "SAUP: Situation Awareness Uncertainty Propagation on LLM Agent". *Annual Meeting of the Association for Computational Linguistics*, 2025.

[11] Xuefeng Du, **Yiyou Sun**, and Yixuan Li. "When and How Does In-Distribution Label Help Out-of-Distribution Detection?" In: *International Conference on Machine Learning*, 2024.

[12] Soumya Ghosal*, **Yiyou Sun***, and Yixuan Li. "How to Overcome Curse-of-Dimensionality for OOD Detection?" In: *AAAI Conference on Artificial Intelligence*, 2024.

[13] Chen Ling, Xujiang Zhao, Xuchao Zhang, Wei Cheng, Yanchi Liu, **Yiyou Sun**, Mika Oishi, Takao Osaki, Katsushi Matsuda, Jie Ji, Guangji Bai, Liang Zhao, and Haifeng Chen. "Uncertainty Decomposition and Quantification for In-Context Learning of Large Language Models". in: *North American Chapter of the Association for Computational Linguistics*, 2024.

[14] Longfei Ma*, **Yiyou Sun***, Kaize Ding, Zemin Liu, and Fei Wu. "Revisiting Score Propagation in Graph Out-of-Distribution Detection". in: *Advances in Neural Information Processing Systems*, 2024.

[15] **Yiyou Sun** and Haifeng Chen. *Large Language Model Augmentation with Definite Finite Automaton.* US Patent App. 19/015,018. 2024.

[16] **Yiyou Sun**, Junjie Hu, Wei Cheng, and Haifeng Chen. "DFA-RAG: Conversational Semantic Router for Large Language Model with Definite Finite Automaton". in: *International Conference on Machine Learning*, 2024.

[17] Cong Zeng, Shengkun Tang, Xianjun Yang, Yuanzhou Chen, **Yiyou Sun**, Yao Li, Haifeng Chen, Wei Cheng, and Dongkuan Xu. "DLAD: Improving Logits-based Detector without Logits from Black-box LLMs". in: *Advances in Neural Information Processing Systems*, 2024.

[18] Jingyang Zhang, Jingkang Yang, Pengyun Wang, Haoqi Wang, Yuxuan Lin, Hongyang Zhang, **Yiyou Sun**, Xuefeng Du, Kaiyang Zhou, Dan Hendrycks, Yixuan Li, and Ziwei Liu. "OpenOOD v1.5: Enhanced Benchmark for Out-of-Distribution Detection". *Data-centric Machine Learning Research*, 2024.

[19] Xujiang Zhao, Wei Cheng, Haifeng Chen, **Yiyou Sun**, and Yanchi Liu. *Uncertainty Decomposition for In-Context Learning of Large Language Models.* US Patent App. 18/977,415. 2024.

[20] Xuefeng Du, **Yiyou Sun**, Xiaojin Zhu, and Yixuan Li. "Dream the Impossible: Outlier Imagination with Diffusion Models". in: *Advances in Neural Information Processing Systems*, 2023.

[21] Yifei Ming, **Yiyou Sun**, Ousmane Dia, and Yixuan Li. "How to Exploit Hyperspherical Embeddings for Out-of-Distribution Detection?" In: *International Conference on Learning Representations*, 2023.

[22] **Yiyou Sun** and Yixuan Li. "OpenCon: Open-world Contrastive Learning". *Transactions on Machine Learning Research*, 2023.

[23] **Yiyou Sun**, Yaojie Liu, Xiaoming Liu, Yixuan Li, and Wen-Sheng Chu. "Rethinking Domain Generalization for Face Anti-spoofing: Separability and Alignment". in: *Computer Vision and Pattern Recognition*, 2023, (**Perfect Review Score**).

[24] **Yiyou Sun**, Zhenmei Shi, and Yixuan Li. "A Graph-Theoretic Framework for Understanding Open-World Representation Learning". in: *Advances in Neural Information Processing Systems*, 2023, (**Spotlight, Top 2%**).

[25] **Yiyou Sun**, Zhenmei Shi, Yingyu Liang, and Yixuan Li. "When and How Does Known Class Help Discover Unknown Ones? Provable Understandings Through Spectral Analysis". in: *International Conference on Machine Learning*, 2023.

[26] Yifei Ming, Ziyang Cai, Jiuxiang Gu, **Yiyou Sun**, Wei Li, and Yixuan Li. "Delving into Out-of-Distribution Detection with Vision-Language Representations". in: *Advances in Neural Information Processing Systems*, 2022.

[27] **Yiyou Sun** and Yixuan Li. "DICE: Leveraging Sparsification for Out-of-Distribution Detection". in: *European Conference on Computer Vision*, 2022.

[28] **Yiyou Sun**, Yifei Ming, Xiaojin Zhu, and Yixuan Li. "Out-of-distribution Detection with Deep Nearest Neighbors". in: *International Conference on Machine Learning*, 2022, (**ICML 2022, #12 cited as of 2025-09**).

[29] Jingkang Yang, Pengyun Wang, Dejian Zou, Zitang Zhou, Kunyuan Ding, Wenxuan Peng, Haoqi Wang, Guangyao Chen, Bo Li, **Yiyou Sun**, Xuefeng Du, Kaiyang Zhou, Wayne Zhang, Dan Hendrycks, Yixuan Li, and Ziwei Liu. "OpenOOD: Benchmarking Generalized Out-of-Distribution Detection". in: *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2022.

[30] **Yiyou Sun**, Chuan Guo, and Yixuan Li. "ReAct: Out-of-distribution Detection With Rectified Activations". in: *Advances in Neural Information Processing Systems*, 2021.

[31] **Yiyou Sun**, Brandon Joseph, Alex Deatsch, Robert Jeraj, and Sean Yang Li. "Localization-based Uncertainty Estimation for Safe Medical Imaging Diagnosis". in: *International Conference on Machine Learning Workshop*, 2021, (**Spotlight**).

[32] Stratos Idreos, Kostas Zoumpatianos, Sai Chatterjee, Wen Qin, Brian Hentschel, Michael Kester, Yiyou Li, **Yiyou Sun**, Wangda Wang, Qizhen Wu, Minqi Yang, and Zichen Zhang. "Learning Data Structure Alchemy". *Bulletin of the IEEE Computer Society Technical Committee on Data Engineering*, 2019.

[33] **Yiyou Sun**, Saaketh Narayan Ravi, and Vikas Singh. "Adaptive Activation Thresholding: Dynamic Routing Type Behavior for Interpretability in Convolutional Neural Networks". in: *International Conference on Computer Vision*, 2019.

[34] Stratos Idreos, Kostas Zoumpatianos, Manos Athanassoulis, Niv Dayan, Brian Hentschel, Michael Kester, Yiyou Li, **Yiyou Sun**, Wangda Wang, Qizhen Wu, Minqi Yang, and Zichen Zhang. "The Periodic Table of Data Structures". *IEEE Data Engineering Bulletin* 41.3, pp. 64–75, 2018.

[35] Bolei Zhou, **Yiyou Sun**, David Bau, and Antonio Torralba. *Revisiting the Importance of Individual Units in CNNS via Ablation*. arXiv:1806.02891. 2018.

[36] Bolei Zhou*, **Yiyou Sun***, David Bau*, and Antonio Torralba. "Interpretable basis decomposition for visual explanation". in: *European Conference on Computer Vision*, pp. 119–134, 2018.

[37] **Yiyou Sun**, Tao Su, and Zichen Tu. "Faster R-CNN Based Autonomous Navigation for Vehicles in Warehouse". in: *Advanced Intelligent Mechatronics*, 2017.